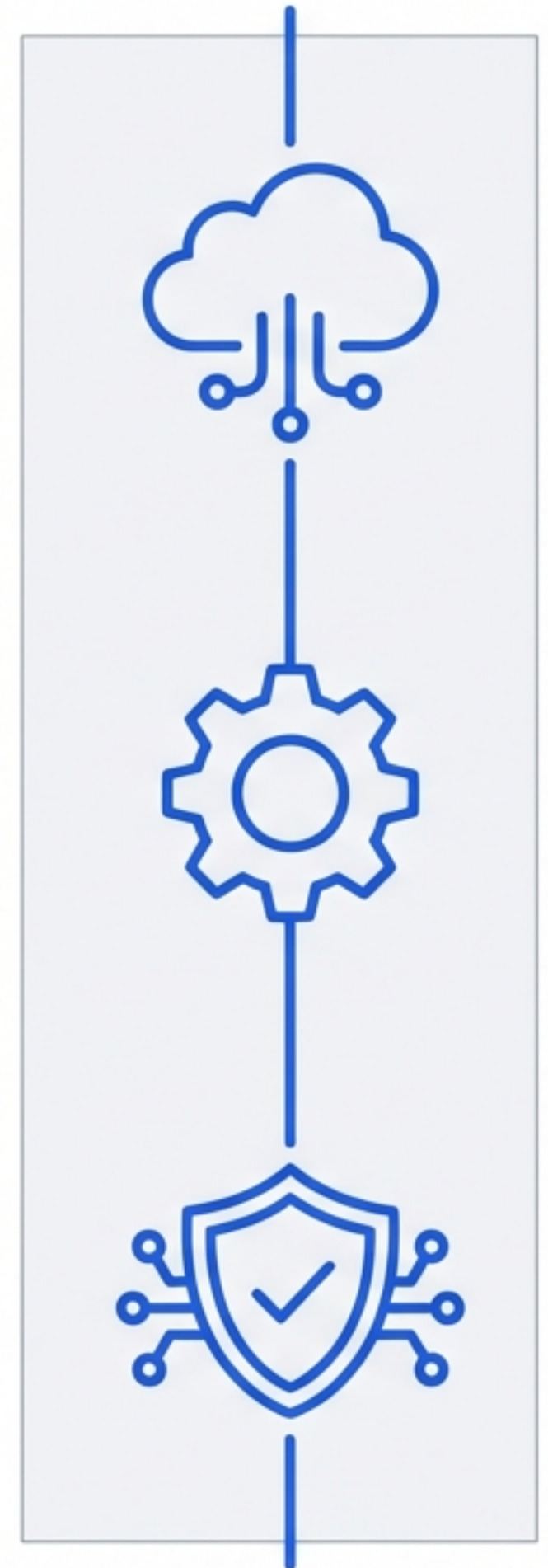


Weather Automation Pipeline

**End-to-End CI/CD Integration
with Python, Docker, & Jenkins**

Author: Marius Nicolae
Repository Status: Public
Context: Technical Portfolio Showcase



Operational Excellence for Routine Tasks

This project transforms a simple utility script into a robust, automated ecosystem. It demonstrates how to containerise, secure, and schedule tasks without manual intervention.



Automated Retrieval

Real-time data fetching via **OpenWeather API**.



Containerised Infrastructure

Jenkins running inside **Docker** for consistent environments.



Secure Delivery

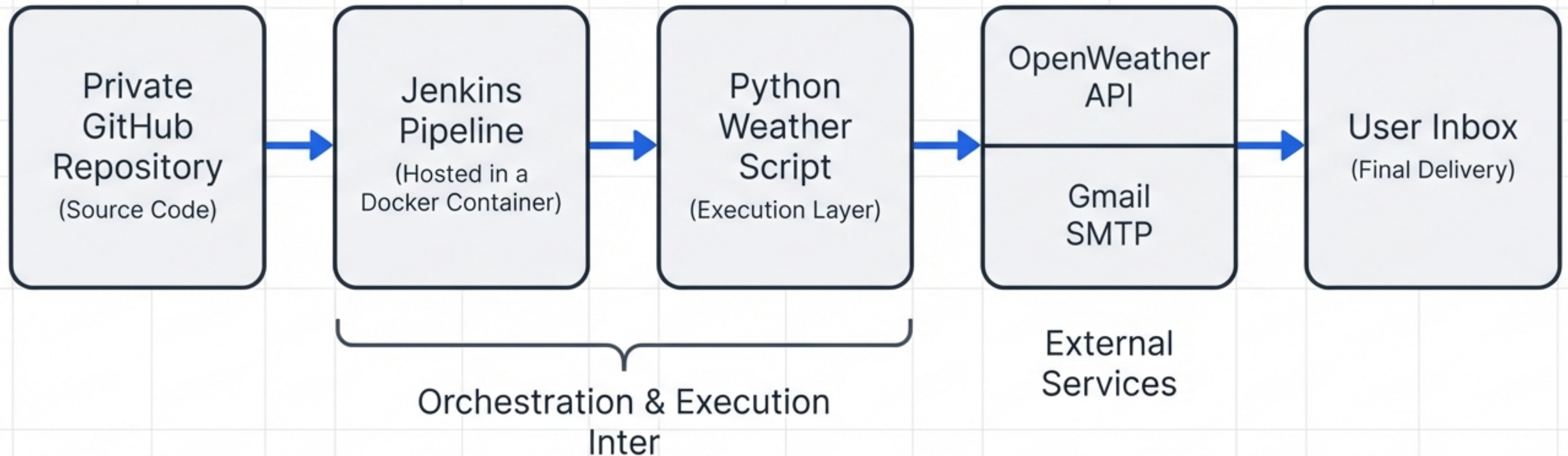
SMTP integration using **Gmail App Passwords** with zero-trust secret management.



Private Integration

Seamless connectivity with private **GitHub** repositories.

From Code Commit to Email Delivery



The Technology Stack

LOGIC

Python 3.13

Weather script logic

ORCHESTRATION

Jenkins Pipeline

Automation workflow

INFRASTRUCTURE

Docker & Nginx

Containerised Jenkins &
Optional Proxy

SERVICES

**OpenWeather &
Gmail**

Data Source & SMTP Delivery

Core Logic: The Python Application

The heart of the pipeline is a concise Python 3.13 script (main.py) designed for modular data processing.

- **Inputs**

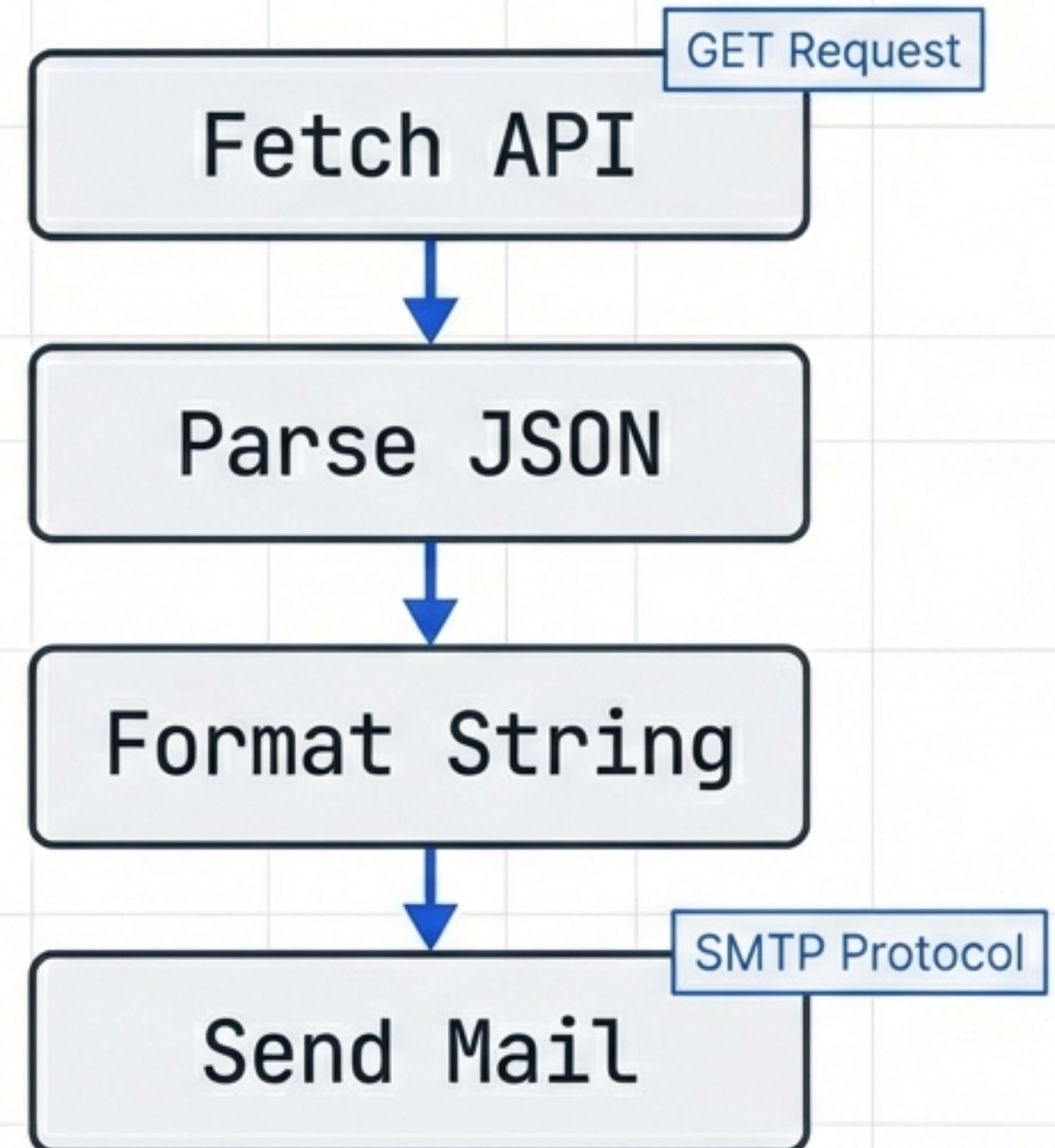
Fetches Temperature (°C), Wind Speed, Atmospheric Pressure, and Humidity.

- **Processing**

Formats raw JSON data into a human-readable summary.

- **Outputs**

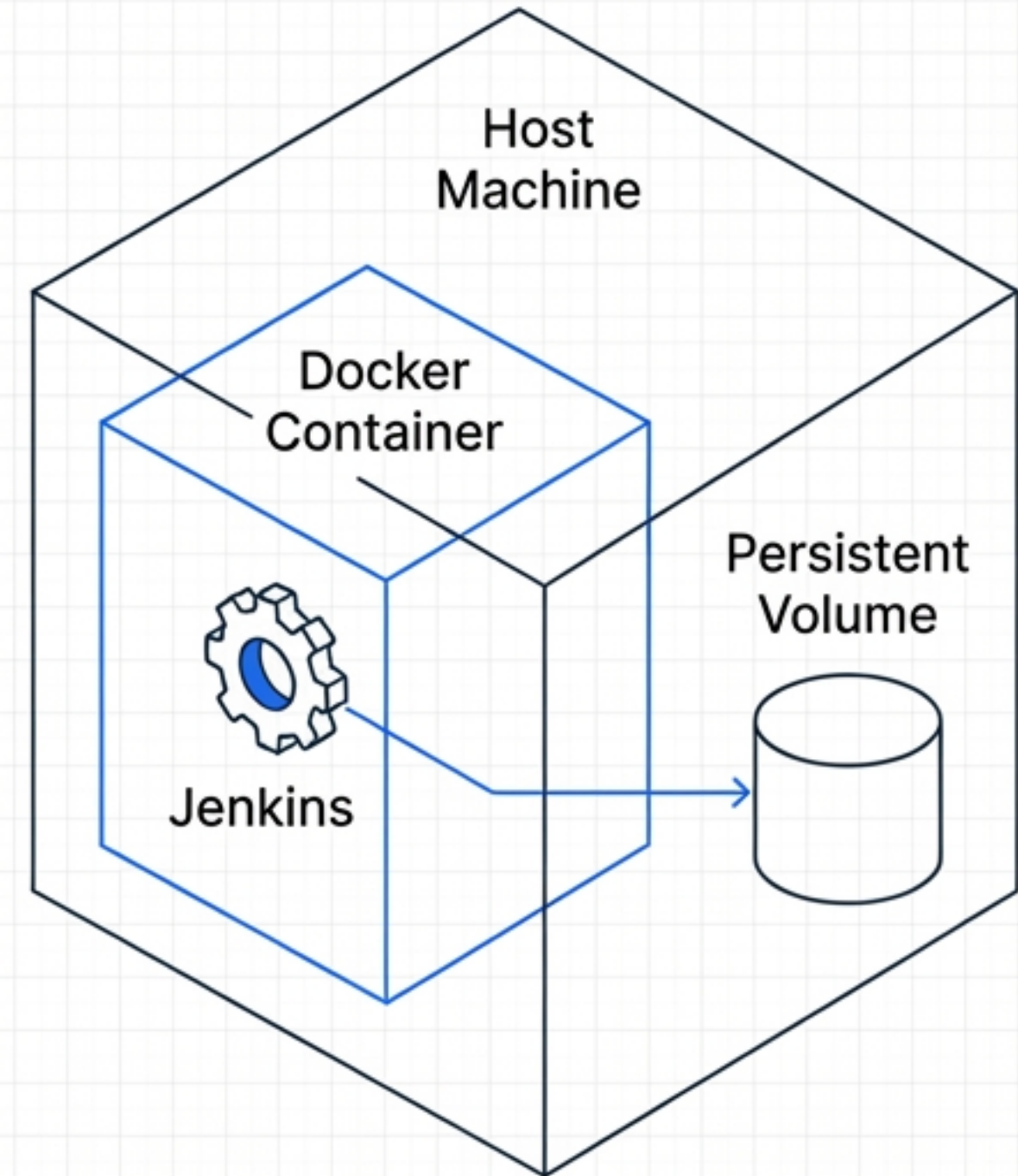
Constructs a payload specifically formatted for SMTP transmission.



Containerised Infrastructure

To ensure the pipeline runs identically in development and production, the orchestration engine is isolated.

- **Docker Integration:** Jenkins is deployed inside a **Docker** container, eliminating 'it works on my machine' issues.
- **Reverse Proxy:** The architecture supports an optional **Ngix** layer for external exposure.
- **Volume Management:** Jenkins-managed **folders** persist data outside the container lifecycle.

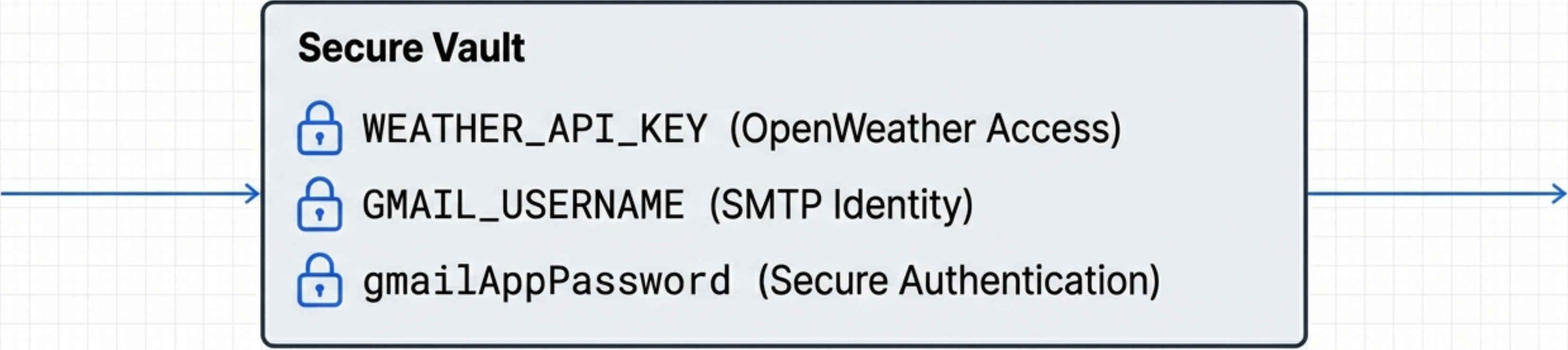


Zero-Trust Secret Management

The Constraint: Hardcoding API keys or passwords in source code is a critical security vulnerability.

The Solution: This pipeline utilises Jenkins Credentials to inject secrets at runtime as environment variables.

Secure Vault

- 
- 🔒 WEATHER_API_KEY (OpenWeather Access)
 - 🔒 GMAIL_USERNAME (SMTP Identity)
 - 🔒 gmailAppPassword (Secure Authentication)

Takeaway: No secrets are ever stored in the GitHub repository.

Orchestration via Jenkins Pipeline

The workflow is defined as code using a JetBrains Mono **'Jenkinsfile'**. This ensures the build process itself is version-controlled and auditable.

01. Checkout



Pulls the latest code from the JetBrains Mono **'Private GitHub Repo'** using a JetBrains Mono **'Personal Access Token (PAT)'**.

02. Environment Setup



Injects the secure credentials defined in the previous slide as environment variables.

03. Execution



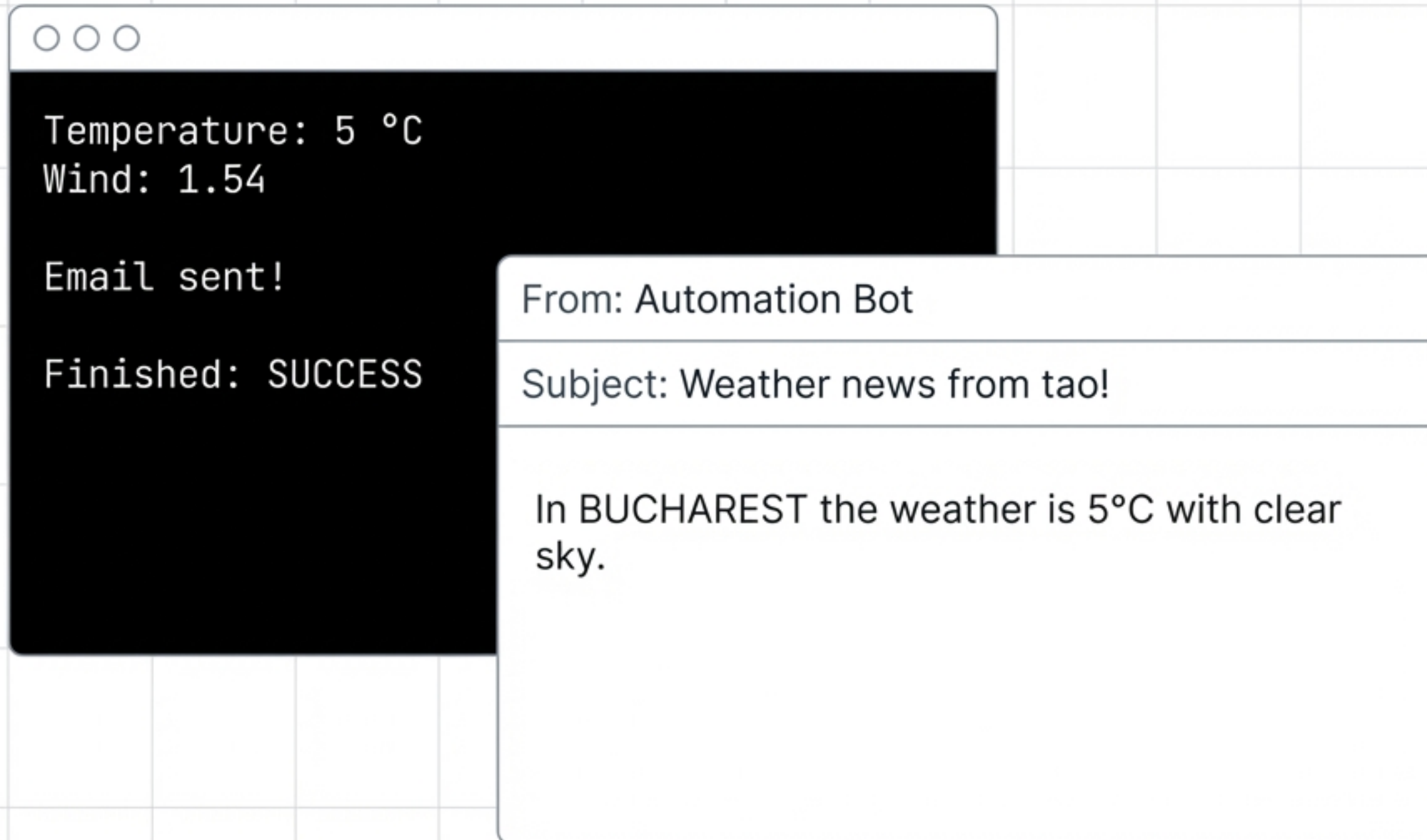
Triggers the JetBrains Mono **'Python'** script within the JetBrains Mono **'Docker'** environment.

04. Reporting



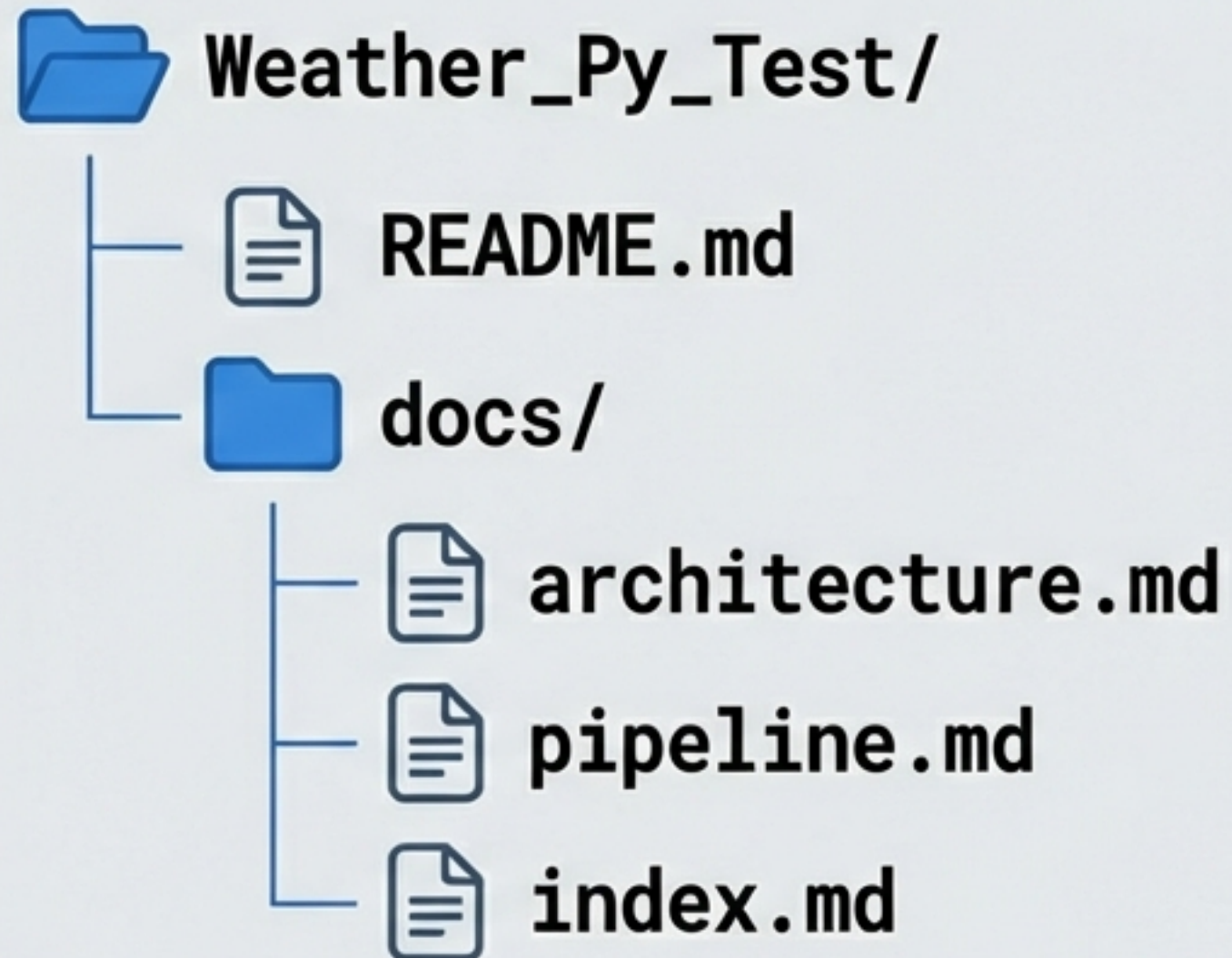
Logs success/failure status to the console.

Delivery and User Experience



Documentation and Reproducibility

The project includes comprehensive documentation hosted via **GitHub Pages**, ensuring easy setup for other developers.



GitHub Pages Setup Path

1. Repo Settings -> Pages
⚙ Path in JetBrains Mono
2. Select Source (/docs or ☒ master)
3. Site is live. ✓

Project Credits & Resources

Marius Nicolae

Automated CI Weather Pipeline



`bytao7mao/Weather_Py_Test`



`/in/nicolae-marius-37b344144/`

A production-grade approach to everyday automation.